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Silver mining monopolies and money relations in the ‘Asian age of the world economy’

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ABSTRACT

The paper argues that Western silver mining monopolies emerged in the Western world in tandem with the Asian age of the world economy. The flow of silver bullion from Western Europe to Asian markets and especially China, was the ‘sustained breadth’ or the macroeconomic condition that mining needed to trigger bundles of money relations at the local and regional levels in dialectical response to such condition. The paper presents a general discussion of two exemplary cases of silver mining monopolies that confirm this argument. In the first case, a local mining corporation in Trent unleashed money relations that expanded both monopoly power over the mines of the city along with monetary relations. The second case, the colonial mining monopolies of Spanish America, shows how money and debt relations increased the speed of mining along with Western European bullion demand in this period. The paper argues that the relation between mining monopolies and money in world history perspective is an urgent line of inquiry to historians and social scientists interested in understanding the centuries-old, dynamic nexus between mining and capital.

1. Introduction

The dominant mining history tradition explains ascent of the silver mining business as a condition built in the causes and effects of the eighteenth-century Industrial Revolution (see for example, Bartels, 2008; Berger and Alexander eds., 2020; Harvey and Press, 1990; Munro, 2015; Vázquez De Prada, 1988). The system of capitalist property and commodity relations embedded in the progress of the Industrial Revolution transformed forms, functions and relations of pre-existing mining ventures into a coherent business structure directly linked to the world economy. Before the eighteenth century, mining, that is, mining outside northern Europe, was isolated from the money and trade relations that pushed accumulation forward. It reproduced ‘pre-modern’ and ‘proto--industrial’ conditions antithetical to capital accumulation (Hilton, 1978; Kriedte, 1992; Molenda, 1988). Low labor productivity, lack of technological change, absence of individual property rights, and above all, production driven by use value, condemned mining ventures outside northern Europe to the ’back seat’ of the economic change triggered by the Industrial Revolution.

This tradition was rooted in the scholarship that for most of the twentieth century, assessed contrasting features between ‘medieval’ and ‘modern’ world by rise of European maritime expansion and the commercial possibilities that came with it. However, this scholarship has long being superseded by new historical inquiries that examine the European world as a subset of a larger, ever-expanding world economy, centered around China, whose territorial expansion preceded that of the Latin, Western Europe, measured absolutely by flows and volumes of trade activity. This scholarship has convincingly demonstrated that East-West monetary and trade relations decided European economic possibilities from the twelfth century until roughly the late eighteenth century (see, for example, Arrighi, 2009; Attman, 1981; Atwell, 1982; Chase-Dunn and Anderson, 2005; Denemark et al., 2000; Frank, 1998; Hodgson, 1993; Kuroda, 2009; Perlin, 1993). More importantly, this Sino-centered world economy reoriented all silver bullion flows of the Eurasian world into China, an economy that increasingly relied upon silver-based moneys from the fourteenth century onwards. In the words of one of the most influential scholars of this new school of thought, the late André Gunder Frank, the ‘Asian age of the world economy’ preconditioned the economic possibilities of the West (Frank, 1998).

The evidence offered by this scholarship is based upon balance of payments, and volume and direction of bullion and commodity trade on a world scale. Its weight and explanatory power call for a reinterpretation of Western silver mining businesses in relation to Asian bullion demand. This paper draws from evidence presented by this scholarship, and departs from the guiding principle that Asian bullion demand was the ‘sustained breadth’ (Braudel and Wallerstein, 2009: 174) of silver

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mining businesses, 'medieval' and 'modern' alike. Certainly, silver mining enterprises flourished by dialectically responding to specific geological and geographic conditions, but these responses were systematically linked to the macro-economics of Asian bullion demand. This paper presents two cases of Western monopoly enterprises that illustrate how mining contributed to the expansion of silver demand during the Asian age of the world economy. The first case is the mining monopoly of a corporation in thirteenth-century Trent, and the second case is the mining monopolies that flourished in colonial Spanish America. In both cases, monopolies guaranteed a safe beginning for the mining business, by doing what captialist monopolies do best: concentrating competition among equals. The money relations embedded in both cases built the necessary 'forward and backward linkages' to local and regional economies, the 'monetary infrastructure' (Perlin, 1993) that expanded bullion demands and supplies of a silver-based Asian age. These mining monopolies allowed underdeveloped Europe to mine its lodes, first, and American lodes, later, in order to have the bullion needed to compete for 'a seat' as Frank puts it (Frank, 1998) in an expanding silver-hungry, Sino-centered world economy.

The aim of this article is to draw the attention to the need to build a global, integrative, macro-history of mining, and one that explains the centuries-long, dynamic relations between mining, money, and capital on a world scale. For doing so, I believe, can allow us to safely move away from 'the narrow loyalty to some sort of ideology' (Heers, 1974: 609) that forces one to fit and reduce explanations about mining capital and profits to the dynamics of change triggered by the English Industrial Revolution. The difficult challenges that the past and present of mining have imposed upon us leave us with no other choice. Did mining precipitate monetization of economic life in extractive resource hinterlands, linking the fate of lodes and miners alike to expansion and contraction of the world economy? Did money relations in mining localities retrench when State debt increased, leaving mining in a state of violence and crisis? Did the 'monetary infrastructure' built by mining monopolies expand or contract the economic possibilities of the West in the Asian age of the world economy? The sections below address in general strokes these questions, by describing two examples of mining monopolies and their most enduring money relations. The article concludes with short ruminations about lines of inquiry on mining monopolies in the 'Asian age of the world economy.'

2. The local mining monopoly of Trent

The silver-based European *grossi*, harmonizing all other currencies used in East-West trade, rose in tandem with a dramatic exploitation of European ore-yielding lands (Bautier 1951; Kuroda, 2009). Silver mining enterprises, flourishing everywhere 'the benevolent hand of Nature' planted ore veins, immediately developed the monopoly forms and functions that fed the dense and rich mosaic of money and trade relations of the times (Braunstein, 1983; Blanchard, 2005; 2001a, 2001b; Ludwig, 1991; Mainoni, 2001; Molenda, 1988; Vázquez De Prada, 1988). In Trent, a young medieval society about to make history, mining also flourished. The city was an ecclesiastical principality, with a social history that incessantly clashed against the nature of feudal law. Its politics and social life were inseparable from the *realpolitik* of a Carolingian Empire on the rise. Nested in an Alpine valley north of Verona, its Monte Calisio was the envy of the powerful Italian city-States that lied in the vast Lombardian plain to the south. For Monte Calisio was the mountain upon which 'the fortunes of Trent were built' (Zammateo, 2016: 48). Known to medieval miners as *mons arçenterie*, this austere mountain stands between Trent and Civezzano. One of the largest watersheds of the Mediterranean region surrounds this mountain (Chiogna et al., 2016: 431; Christie, 2006: 358; Franceschini, 2016: 251; Perini, 1856: 175).

The Adige River capriciously played with the fate of lands and peoples alike. It remained for most of its history a *rapidissimo fuime* prone to floods (Giulio, 1845). In the times of Bishop Frederick Wanga, the Sovereign lord who compiled the first mining law of the city in A.D. 1207, the Adige River expanded into neighboring streams several times, swallowing agrarian lands and impacting underground water levels (Basetti e Borzatto, 2005; Castagnetti, 1982). The Alps did then what still do best: they added more water to the valley, creating seasonal changes in surface and underground waters (Brázdil et al., 1999: 266; Wehren et al., 2010: 50). Alpine snowmelt increased groundwater storage in the watershed of the river, leading to floods that turned improved lands into swamps. These hydrological interactions incessantly impacted the water table of mines (Giulio, 1845: 116; Hartungen, 2013; 67–68).

Competing land uses built here led to continual and ferocious competition among the ruling classes, who sought to grab higher shares of agrarian and mining surpluses (Archetti, 2011: 487; Franceschini, 2016: 257; Varanini and Demo, 2012: 279). Silver mines were especially coveted, as ores were needed for satisfying increasing demand for bullion. When Bishop Wanga came to power, due to family relations with the well-connected Count of Tyrol, Monte Calisio mines needed draining (Casagrande, 2014; Tizzoni, 1985). Documentary evidence in the *Codex Wangianus* (A.D. 1185/1206) unfortunately says nothing about the specific event that flooded the mines. We know, however, that in response to the problem of shaft flooding, Monte Calisio became a mine of multi-level shafts connected by fortified tunnels (Tizzoni, 1985). This was mine geometry that required money to pay for infrastructure and labor costs (Franceschini, 2016; Gorini, 1986). An experienced foreman most likely supervised drainage operations, planning for infrastructure work, above all, the installation and operation of *rotam* or waterwheels adapted to mining and smelting purposes (Reynolds, 1983). He was a miner familiar with general geological and geographic conditions of the mine. This experienced foreman was also knowledgeable about prices and local market conditions for the raw materials needed for improving infrastructure. The latter included different varieties of timber, turf, charcoal, animal hides, leather sacks and bags, and buckets. Above all, he was familiar with the workings of waterwheels and the costs of repairing or rebuilding these machines.

Ore extraction increased the demand for money locally, because it required tools bought in local markets. These local markets were dominated by the bustling activity of blacksmiths, carpenters, artisans, and farmers. The tools needed included pickaxes and hammers of different sizes, buckets and leather bags for carrying and storing water, and ladders. Most likely, many tools were redesigned and adapted in situ, increasing the mining bill, as materials had to be available in the mining site for carpenters and blacksmiths to do their job. Because ladders were needed, the cost of transport of timber also added to the bill. Mine illumination also increased the need for money. Archeologists digging Monte Calisio have discovered short and irregular shelves carved out of the rockwalls and used for placing wax- and resin-candles and oil lamps, suggesting an extra money linkage between mining and the local economy (Casagrande et al., 2013: 189; Zammateo, 2016).

The highest demand for money came from ore processing. Smelting and cupellation were the silver-recuperation techniques *par excellence*, as discussed extensively elsewhere (Claughton, 1993; Martiñón-Torres, 2008; Rippon et al., 2015; Vázquez De Prada, 1988). Smelting separated the unwanted materials from the ore vein. It required smelting the silver-bearing lead ores and 'de-silverization' of lead bullion and recuperation of silver (Tizzoni, 1985: 285–289). Cupellation required 'converting lead to litharge or lead oxide before it could be separated from silver' (Rippon et al., 2015: 98). It was a 'high-temperature metallurgical operation aimed at refining the noble metals' (Martiñón-Torres et al., 2008: 60). Boles and bellows-blown furnaces were employed in this first step, depending upon size of ores, among other conditions, and cupels were used in cupellation (Rippon et al., 2015: 89).

The real problem with the forward and backward money linkages described above was that expansion of East-West commerce led to money depreciation in Europe relative to Asia. This macroeconomic condition had tremendous implications at the local level. Lords experienced this problem as the 'falling rate of feudal levy,' a condition aggravated by the fact that income grew faster than population in the first silver century of the European economy (Cipolla, 1963: 417). The speed of income expansion slowed down up to the middle of the thirteenth century, peaked again until the beginning of the fourteenth century, and slowed down again for the rest of the fourteenth century (Cipolla, 1963: 418). Lords responded by introducing monetary payment systems complementary to coined silver-based moneys in order to solve their need for 'silver cash' (Bautier, 1951; Saccocci, 2010). By no means this was a concerted, conscious response to the macro-economic forces at stake. But conscious or not, lords began collecting tithes in agrarian commodities, valued in the silver moneys in demand. The *gastaldia* or tax collectors of Bishop Wanga collected cheese, *vaccas & porcos*, rams and lambs, and wine and oil a spart of the taxes owed to the lord. They adjusted feudal taxation to cyclical agrarian activity, effectively putting into practice a feudal accounting system that transcended the problem of low supply of silver bullion for coinage purposes (Curzel and Varanini, 2011: 573–574; Schneller, 1898: 161). Lords also decreased silver content of coinage, to increase the number of currencies in circulation, increasing money in circulation in small cities like Trent, where small coin denominations mattered greatly, even if debased (Spufford, 1988). The lord Bishops enjoyed after all, coinage and minting rights since A.D. 1182 (Gorini, 1986: 240; Kink, 1852: 432). They minted silver *grosso* until A.D. 1220 (Gorini, 1986: 241).

Under these conditions, granting private producers a monopoly of mining rights was the only developmental path available to lords. From the perspective of lords, mining required not just money, plain and simple, but money ready to be *regularly* ploughed into land (Castagnetti, 1982: 27–29; Malfatti, 1883: 17; Voltellini, 1981: 23). Hydrological conditions of Trent so demanded. From the perspective of the private producers, the question of money was sligthly different. It was not just that they owned ready cash to be ploughed into land regularly, but if the ploughing of money could into land could yield more. Only a monopoly promised to achieve both ends, even when unleashing economic clashes between lords and miners.

The extensive mining contract granted by Bishop Wanga in A.D. 1208, the corporation or *consilium* that drained the mines of Monte Calisio enjoyed monopoly rights. Unlike the typical *ars* or guilds that flourished in Florence and other Italian city-States, the *consilium* had a rigid, vertical hierarchy. At the top stood the group of *silbrarii*, members of the corporation and owners of mining shares (Curzel and Varanini, 2011: 352). No other economic agent linked to *consilium* enjoyed ownership rights over mining shares. This condition indicates that only *silbrarii* enjoyed the gains of monopoly mining capital. The lord Bishop was denied the advantage of owning shares, suggesting intriguing questions about the relation between money and politics. This situation contrasted with a mining contract granted in A.D. 1185, in which a *gastaldia* working on behalf of the then bishop joined two other members of an early association holding mining rights. However, Bishop Wanga but appointed his most trusted noblemen as tax collectors who exercised general oversight of mining and drainage works and were thus familiar with the business of *silbrarii*. These officers had the responsibility of enforcing certain regulations of the mining contract. Because no other economic agent of the medieval economy enjoyed ownership rights, only *silbrarii* accumulated any gains to be made from the mining monopoly. To enjoy any gains, each *silbrarii* had the obligation of paying for his share of mining expenses, including infrastructure repairs, mine improvements, and smelting works, on a weekly basis.

Next in the hierarchy were chief engineers, who embodied other money relations as they were paid in coined money and crude ores. They were responsible for deciding over matters pertaining mine geometry and infrastructure. They belonged to a class of highly skilled miners, identified as *omni werki* or free miners from Germanic origins (Graulau, 2019). They were responsible for ensuring that mining works ran according to plan, especially during the season of high hydrological activity. It is possible that *silbrarii* and *werki* were one and the same in this period, when general technological conditions of the European economy left little room for differentiation between producer, technical expert, and scientist, as it did in the nineteenth century (Allen, 2017: 73, 120). Below engineers stood a diverse group of miners and laborers hired for sinking shafts, digging tunnels, and extracting ores. They possessed different degrees of specialization and skills and occupied the lowest stratum of the labor hierarchy. At the bottom of the hierarchy stood a diverse group of workmen (miners, shovelers, windlass operators, ore washers and carriers, smelters) performing manual tasks. Many of these *laboratores* were 'employees' hired on a short-term basis and were dropped and added when needed. They endured the heaviest toil and trouble of mining works with little monetary remuneration. They were, nonetheless, integrated into the local circuits of silver money, but theirs was a subordinated integration given their labor situation. Miners worked the shafts for five to six days per week, in two daily shifts, each of eight-hours duration, as was the practice in other mining districts (Graulau, 2019).

The money relations embedded in the business of mining shares also reveal the linkages between mining and the wider world of bullion demand. The *consilium* divided each mine into sixteen shares or *partem*, each share representing a fixed amount of money invested by *silbrarii* in a specific shaft or section of a mine. *Silbrarii* thus participated in regional bullion markets, because they set prices and values of shares considering costs of mining works as well as expected gains. Calculated on a weekly basis, the cost of all sixteen shares equaled the total expenditure in mining works undertaken by the *consilium wercorum* (Graulau, 2019). If a claim holder neglected his duty of investing money regularly in mining works, while the rest of the members of the corporation paid total expenses of mining works, that shareholder was responsible for reimbursing expenses to the rest of the corporation. Failure to spend money could end participation of *silbrarii* in the *consilium*. The contract established that a fifteen-days period was the deadline for a shareholder to bring his shaft into production and settle debts with the rest of the corporation (Curzel and Varanini, 2011: 352). If a shareholder was not able to pay off, nor reimburse the members of the *consilium* in a period of fifteen days, including payments owed to technical specialists and workers, his share 'in total' was transferred to other members of the same *consilium*. A shareholder could rescind from participating in further investments in a fifteen-days period but conditioned to his paying his share of mine improvement works. Above all, mining shares functioned as security and could be bought and sold among *silbrarii* (Mainoni, 2001: 435). Share prices were based upon gains and expenditures of the mining claim including expenditures lost to the mine or irreparable damages to infrastructure and improvement works. Shares thus broadened demand and circulation of bullion, allowing *silbrarii* to maneuver the inflationary tendencies of the economy, with political consequences that we cannot fully address here. Suffice to add that the *consilium* became a moneylender to the lord Bishop of Trent, conceding a series of loans that most likely were repaid with the ore bunties of Monte Calisio.

The lord Bishop participated nonetheless absorbed some of the surpluses extracted by the consilium from Monte Calisio mines and miners. He imposed taxes and fines against violations of the mining law of Trent. The contract established a fine of twenty-five pounds against *silbrarii* failing to declare total mining gains or obstructing the oversight role of the *gastaldia*. It also established a heavy fine of 50 pounds against those damaging drainage works. Fines were levied against illegitimate ore sales, bringing weapons inside mines, setting mine works on fire 'maliciously,' and others (Graulau, 2019). *Gastaldia* also collected from the *consilium* the mineral *fictum*, an amount of bullion due to the lord and based upon actual mine accounts of the *consilium*. Refusing the payment of the *fictum* to the *gastaldia* carried a heavy fine: *in duplum ficto*, an additional amount of twenty pounds (Curzel and Varanini, 2011: 352).

The business of the *consilium* continued for some years, based upon a contract that appeared five years later. Then, the lord Bishop extended legal protection to the monopoly held by the *consilium*, prohibiting miners from carrying on work that obstructed the drainage adit (Graulau, 2019). The new contract also created a mining tribunal composed of technical experts, whose duties included settling controversies among owners of mining claims in cases when a shaft was sunk in the direction of a drainage adit. The *consilium* by then became a moneylender to Bishop Wanga, tightening the relations between their silver mining venture and the expansion of commerce and trade of this silver century.

Most likely, the bullion produced in Trent found its way to Mediterranean commercial cities, in route to Islamic commercial cities and Central Asia, as European bullion did in this period (Elbl, 2007; Saccocci, 2010; Spufford, 1988; Steuer, 2004). The rise of a silver-based monetary payment system in China decisively impacted the fate of mining in the Western after the mid-fourteenth century (Flynn, 2019; Kuroda, 2009; Frank, 1998). In the short run, it added to the inflationary pressures of the European economy (Flynn, 2019: 180). We hear nothing of the *consilium* in the fourteenth century. By then, silver mining businesses had flourished in Sardinia, Massa Marittima, and Bohemia, until displaced by silver mining enterprises of Spanish America two centuries later. The latter became the epicenter of an unprecedented production of world liquidity, as described below.

3. Colonial mining monopolies in Spanish America

The expansion of Spain and Portugal in the fifteenth and sixteenth centuries came along a rapid commercialization and monetization of economic life everywhere. For the first time, a silver-based currency, the Spanish *real de a ocho* or Spanish *peso*, reached all regions, creating balances and imbalances of trade on its way its 'end-market' in China (Flynn, 2019). From the banking centers of Europe to the low-end markets in rural hinterlands of Spanish America, silver-based moneys were in demand. Silver mining enterprises that flourished in Spanish American colonies were at the center of this development (Bakewell, 1971; Brading and Cross, 1972; Craig and West, eds., 1994; Gil Montero, 2014).

Silver mining in Spanish America was the offspring of wars of conquest by the Spanish Empire first, later joined by Portugal, England, France and Holland. Spain, however, was the territorial power that pioneered the unprecedented wave of mining colonization that took place after the conquest of the Mexica and Inca empires. Mining fiercely reproduced monopoly relations right from the start. The Castilian Crown (later to be the Spanish Crown) claimed sovereignty over all American lands immediately after the first voyage of Columbus. It was a consolidated monarchy that sustained its imperialist foreign policy by exploiting monopoly profits, legitimized by fiscal and legal instruments building the doctrine of Sovereign *Iura Regalia*. Before the conquest of the NewWorld, Castille paid for its crusading wars against Islamic Andalusia and northern Africa with revenues stemming from monopolies of agrarian production and trade. It enjoyed mineral rents from silver, gold, and salt mines located in its territory. Castile also extracted surplus from duties such as the *alcabala* and *almojarifazgo* applied to imported commodities, and taxes to Jewish communities of the kingdom (Canto García, and Cressier, eds., 2008; Echevarría Arsuaga, 2010; Ladero Quesada, 2011: 96–97). As its territorial and maritime expansion progressed, Castile adapted its fiscal instruments to novel economic possibilities it encountered in the colonies. Upon Columbus return to Spain from his first voyage in A.D. 1493, Castile claimed its *Iura regalia* over all lands and mines, 'discovered and to be discovered,' in the New World (Elliot, 2007). Such was a propitious juridical principle for tying the fate of mining ventures to the monetary relations of the Crown.

Spanish conquistadors opened some of the most remarkable mines in Mexico, then the Viceroy of New Spain, and Bolivia, then part of the Kingdom of Peru (Bargalló, 1955; Lacueva Muñoz, 2010; Miranda Díaz, 2014; Serrano Bravo, 2004). Rights took off as *asientos de minas*, understood as rights to open and benefit from mining works in lands containing ores and conquered Amerindians. These were individual rights, granted given to individual conquistadors, not corporations, as rewards 'justly' earned for conquering lands, pacifying Amerindians, and expanding the colonial frontier of the Empire. Zacatecas and Potosi were, however, possibly the most famous *asientos* that took bullion production and circulation to unprecedent levels. The ore veins of Zacatecas were hidden below 'gently rolling hills' on the Western edge of the central plateau of Mexico (De Cserna, 1976: 1191). The land here is part of the semidesert region of northern Mexico, abundant in several varieties of cacti including agave. Water here was supplied by wells and small, narrow streams. Miners exploited first the famous Fresnillo veins, which produced 12 tons of gold, 775 tons of silver, and one million tons of combined lead and zinc, from their opening until the mid-1970s. Miners sunk different shafts connected by tunnels, that in time lead to even richer veins. A mine claim was legitimate if it measured approximately 110 m long and 55 m wide on surface land (Brading, 1971: 131). A miner had 20 days to register the stake after discovery, suggesting problems with location of mines. Humboldt, writing in the eighteenth century, stated that '100 leagues' separated Zacatecas from Mexico City (Humboldt, 1822: 140). The road was only transitable by horse or mule pack. It was dangerous to tax collectors and merchants, who continually complained about the *indios de guerra*, or groups of Amerindians who raided trade convoys.

Once the stake was registered, the law granted miners a four-month period to show evidence of continuous operation (Molina Martínez, 1998: 1020). Generally, miners dug a principal shaft reaching up to 300 feet below surface land, perpendicular to surface land. Connected to this shaft, narrow drifts dug at different levels formed *labores de chiflón* or interconnected passages (Semo, 1993). Props placed for securing weak sections of a wall known as z*apatillas* added strength to the structure. Rock-cutting; ore-transporting; feeding and tending for beasts of burden; removing waste rock from shafts; keeping mine entrances clean; and all other burdensome and strenuous tasks were done by Indian labor force. Fire-setting for separating ore veins from rocks, when the technique was employed, was done by slaves of African origins. While Zacatecas ores were mined at an altitude of 3000 feet above mean sea level, Potosi miners had to reach the clouds in order to mine the ores, in a manner of speaking. The mountain that contained all mines stands at 13,200 feet above mean sea level and has a 'steeply conical summit' (Brading and Cross, 1972). The city lies at its feet, surrendering its economic and social life to the mining economy. Both share the Andean barren soil of the region and a cold climate. This remarkable geography imposed unique monetary functions and relations to the mining business. Horses, ideal beasts of burden in Zacatecas, were of very limited use in Potosi. Here, the Andean llama became the preferred beast of burden (Arzáns de Orsúa y Vela, 1965; Elliot, 2007). Llamas were raised and bred for this purpose, adding profitable agrarian and livestock functions to mining, and from which clever miners increased their gains. Unlike the ores mined at Zacatecas, the rich silver ores of Potosi were found in tightly packed ore veins concentrated in the top hundred feet of the mountain summit (Brading and Cross, 1972). Below this bonanza, lay even greater ore veins but were more dispersed, until they reached the bottom of the mountain, where ore veins were high-grade sulphides, with veins of native silver or argentite, even richer than those at the top.

Miners dug open pits in various zones of the mountain, following the richest and more concentrated veins. They extracted ore-containing rock using iron bars, hammers, wedges, and other tools made of stone, iron, copper and bronze. Extraction shafts were also dug, reaching as much as 70 m in length and connected by very narrow tunnels. These were so deep that it took one entire day shift for a miner to transport rocks outside the mountain (Serrano Bravo, 2004: 19). Because of the conical shape of the mountain, adits became extraction shafts, a situation that in time lead to the problem of erosion and soil collapse. A miner had 60 consecutive days to open and 'populate' a mine, revealing a more challenging location of mines. The distance between the mountain and the nearest sea was 643 kms, covered by mine inputs and outputs. It possibly added more kilometers 'with each pack because of the necessity of following pasture lands' (Cobb, 1949: 25). This condition, as in

Zacatecas, increased the cost of mining, and thus, the prospects of making profits from loans and credit relations.

The *patio* process was the silver-recuperation method used in the colonies. It was the greatest metallurgical gift of the Americas to the rest of the world, an done that deepened forward and backward linkages between mining and the rest of the colonial economy. Invented by Bartolomé de Medina in A.D. 1554 while working at the Purísima Grande mine in Pachuca, it was a lengthy process for cleaning low-grade silver ores by mercury amalgamation (Castillo Martos, 2008). The first stage was to place the ore-bearing rocks in a large, covered box, and crush and pound the box with wooden crushers or *morteros* with heavy iron pestles at the end, in a stamp mill. Operated by horsepower, the machine consisted of an axle to which the arms of each crusher connected. After continuous pounding, the small pieces of ore fell upon a sieve made of hide, separating large pieces form smaller ones. After a second pounding, the ore was transferred to an *arrastre*, or circular ore-grinding mill moved by mules and placed on even ground level. The ore was pulverized into fine dust by using heavy and hard granite stones, connected with a revolving shaft operated by animal power. The miner added water while pulverizing the ore, forming a muddy mass. This mass was placed onto a *patio* or floor made of hard stone. Generally, 2 1/2 to five pounds of salt, 5–10 pounds of copper pyrites, and 15–20 pounds of local lime were added per hundredweight of mass. Mercury was added in proportion to the estimated silver yield of the crude ore. The general proportion was 3–4 pounds of mercury per mark of silver (Brading, 1971: 138).

When everything was uniformly mixed, the result was a strong brine or *caldo*. This *caldo* was separated into piles weighing between 15 and 32 hundredweight. The mixture was left in the open air, exposed to sun heat, and stirred daily by men and horses tramping over it. This stage could take between two and eight weeks, and depending upon the smelter's judgement, copper pyrites, salt and mercury were added. The purpose was to create a 'cake' or *torta de lama*, where mercury, salt and muddy ores were all uniformly mixed. The miner deposited the 'cake' in vats for thorough washing, leaving the *plata piña* or amalgamated silver. This amalgam was placed into canvas bags and submitted to heavy pressure to extract the unwanted mercury. Finally, the mixture was placed in an oven fueled by charcoal, where a chemical reaction occurred whereby silver was separated from mercury, and silver was collected from the bottom of the oven. Alonso Barba adapted the entire *patio* process to the nature of Potosi ores (Barba, 1729). His *cazo* method applied heat to the muddy mass of ore, after mixing the mass with mercury and salt (Semo, 1993).

The *patio* process became an engine that intensified monetization of daily mining life. *Morteros* were bought and imported from regional markets. When brought from Spain, they were assembled in the mines. Horse, originally brought from Europe, were bought in regional *haciendas* that took off in the semi-arid lands of northern Mexico. Zacatecas miners purchased salt mined at the *salinas* of northern Mexico and the Yucatan peninsula. They obtained charcoal from timber coming from the forests of northeast Mexico, leading to deforestation that mining laws addressed much later (Molina Martínez 1998). In Potosi, miners used the endemic *icho* straw as fuel for ore processing (Cobb, 1949: 26). Salt, however, was a Crown monopoly, obtained from tribute imposed unto Amerindian towns located on the Pacific seashore. This sea salt came in two varieties (*blanca* and *bermeja*), both sold at public auction to Potosi rich miners, and merchants (Borah, 1992: 69). A second source of salt were the *salinas* for which Bolivia is famous today. The large salt lakes were possibly mined for salt by Amerindians, pulled into the functions of the mining economy by colonia laws. This salt was less pure for the *patio* process, and when used, the miner had to make smaller adaptations to the process of ore smelting,

Mercury travelled across the Atlantic from Almaden and Guadalcanal mines, in Seville. It was a monopoly held by the Spanish Crown (Ramos Pérez, 1970). The *Cajas Reales* or regional treasury offices distributed in the colonies sold mercury at 100 marks of silver per hundredweight. The high price strengthened monopolistic competition among the rich miners of the colony (Brading, 1971: 142). Overland transport of mercury, from the port of entry at Vera Cruz in the Yucatan Peninsula, to all mining districts to the north of Mexico City, was a route guarded by colonial authorities. This inland route was known as *Camino de Europa* or *Camino de los Virreyes* (González Tascón, 1997: 1301–31). Later mercury was supplied by the mines of Huancavelica, in Peru, which by relied upon the labor of 4000 Amerindians, according to a contemporary observer (De Baeza, 1607: 9; Lohmann Villena, 1997: 176).

The money relations described above sustained the rigid social hierarchy and division of labor that characterized colonial mining. Spaniards and other Europeans occupied the top of the mining hierarchy. The middle stratum of the social hierarchy was composed of a mixed class of small-scale entrepreneurs, traders, blacksmiths, experienced miners who were paid with salaries, and artisans. The lowest stratum was composed of a mosaic of free and semi-free forms of labor, and slaves of African origins, dominated by Amerindian labor. Amerindian labor was historically rooted in *encomiendas*, a military-religious institution that Spain adapted to the conquest and colonization of Amerindians and their lands. This was a land grant to conquistadors who participated in conquest wars. The land grant presupposed the responsibility of conversion of conquered peoples into the Catholic faith. It truly meant that conquistadors had a captive labor force to improve and develop the lands of the colonies (Elliot, 2007). *Encomiendas* allowed Spaniards to extract tribute from conquered Amerindians. The problem was the high mortality rates of *encomiendas*, a problem that threatened the production of bullion needed back in Spain (Gil Montero, 2014). In A.D. 1542, changes to *encomienda* laws introduced the obligation of owners of mines to compensate Amerindian labor with a *jornal* or salary, paid in money (Assadourian, 1978; Semo, 1993). The 'social justice' impact of the Amerindian right to a salary change was minimal, as expected, as colonial structures severed already broken communal Amerindian relations. In fact, this regulation legitimized the transformation of the bulk of Amerindian peoples into a reserve army of labor (Semo, 1993).

The Potosi *mita* guaranteed surplus extraction from forced Amerindian labor, analogous to the *encomienda* system (Miranda Díaz, 2014). It was an 'Indian wage-labor system' whereby the Amerindians received monetary compensation for the hours spent working in the silver mine. It was a labor-service obligation that allowed European miners to rely upon a steady supply of Indian labor force. Potosi received annually between thirteen and sixteen percent of the male adult population of the towns (Brading, 1971: 103). At any month of the year, Potosi had between 4000 and 5000 active *mitayos* working in the mines (Elliot, 2007). Amerindians were legally 'free Indians' and entitled to receive a salary in compensation for providing labor to the mines (Gil Montero, 2014: 7). Such a salary, when paid, linked the lowest stratum of the mining economy to the wider colonial economy, as Amerindians were forced to pay taxes in money. Moreover, Amerindians had to sell their labor to mine owners and incur debt to pay their share of taxes (Gil Montero, 2014; Hernández Sobrino, 1999).

The mines of Zacatecas and Potosi were centers of world liquidity (Frank, 1998). Only the output of the mines of Zacatecas registered at the Spanish Royal Treasury was 545,358 Marks in A.D. 1559–1563; 664, 112 in A.D. 1564–1568; and 743,868 in the next five years. By comparison, the entire European silver-mining industry during produced 350, 000 Marks (of 8½ ounces) a year during the same period (Bakewell, 1971: 259–270). At Potosi, 'production was seven times higher in 1585 than in 1572, and during the period 1580-1650 output never fell below 7.6 million pesos' (Kamen, 2003: 286). By the end of the sixteenth century, the output was higher than that of the entire European silver-mining industry. The immediate impact of this liquidity bonanza was inflation (Flynn and Giráldez, 2010). Availability and scarcity of the metallic currency the world economy demanded, the Spanish *real de a ocho* or Spanish *peso*, led Europe into massive trade imbalances with silver-buying China. 'Between 1600 and 1800, continental Asia absorbed at least 32,000 tons of silver from the Americas via Europe, 3000 tons via Manila,' and other quantities from the rest of the silver-producing world (Frank, 1998: 146).

Meanwhile, the Royal Treasury was depleted, as the empire alleviated its negative balance of trade by increasing debt. Credit guaranteed access to Asian commodities (Frank, 1998; 150). The Royal Treasury disbursed payments to bankers and creditors in gold and especially silver bullion (Álvarez Nogal, 1997: 21; Herrero Sánchez, 2005; Pike, 1962). This bullion was collected as *el quinto* or the one fifth paid by miners in the colonies. Genoese bankers accumulated 76 percent of all gold and silver from American colonies that reached the port of Seville during A.D. 1621–1626. German bankers, among whom the *Fúcares* or Fuggers figured prominently, received 26 percent in the same period (Carande y Thovar, 1945; Kellenbenz, 2000). This sum was mostly for the payment owed for their supplies of *azogue* or cinnabar to the Royal Treasury (Álvarez Nogal, 1997: 23).

The financial crisis that the Spanish Crown suffered, forcing a series of suspension of payments from A.D. 1536 onwards by no means diminished the monetization and commercialization that mining monopolies entailed (Kamen, 2003). If anything, mining monopolies reproduced debt relations, in tandem with European dependency upon credit as a way of handling the bullion demands of the Asian age of the world economy. In Zacatecas especially, the micro-economics of debt were organized around the *aviador*, the *rescatador* and the *fiador* (Langue, 1992: 70–74). The *aviador* was a banker who was also a miner, or had experience in mining affairs, who advanced loans to miners to pay for mercury supplies and cancelation of debts related to mining operations. *Rescatadores* bought crude ores from small-scale miners who could not afford the costs of the taxes imposed by the Treasury or the prices of mercury. They paid miners without delay and in cash or credit. Sometimes, *rescatadores* owned their own ore-smelting facilities, and thus, enjoyed direct relations with the *Cajas Reales*, to whom they sometimes sold bullion. *Fiadores* were guarantors who possibly handled big money, given that their sphere of action was directly related to financing mercury distribution. These debt relations allowed the surplus of monopolies to trickle down to emerging commercial agents of the colonies.

World bullion demand expanded in the seventeenth century, and prices of Asian 'luxury' goods arriving at Spanish ports increased. The Spanish Empire responded by extending privileges aiming at renewing mining production. New compilations of mining laws emerged, as an attempt by the State to harmonize new social and political conditions impinging upon the silver mining ventures of the colonies discussed elsewhere (Kamen, 2003). By then, the monopoly form of the silver-mining ventures paid off, locally and globally. At the local level, many of the early holders of *asientos* became rich miners, remembered today as their names gallantly adorn streets and plazas in many countries. Francisco de Ibarra, the Basque conquistador who founded modern-day Durango, owned the *asiento de minas* of Avino, in Zacatecas. The settled land of such *asiento* had 300 Spaniards, 500 slaves, 1000 horses and mules, one vicar, and one priest. His mines were *las más ricas de plata que hay en este reino* (López de Velasco, 1894: 137). When the Spanish Empire eased nationality restrictions in mid-sixteenth century, a few rich foreigners became owners of *asientos de minas* in exchange for much-needed loans. The Fugger and the Welser obtained *asientos de minas* for lands located in modern-day Venezuela, as rewards for financing the election of Charles V (Kellenbenz, 2000).

Globally, Spanish American mines allowed the Asian age of the world economy to count on massive amounts of bullion, more so than European mines. Spanish American mines yielded 80 percent of the total world supply of silver and 70 percent of gold until 1800 (Kamen, 2003: 285). Only until the last decades of the seventeenth century, when the fate of the Spanish Empire was already decided, the mines provided between 25,000 and 30,000 tons of silver; later, that amount 'more than doubled' (Garner, 1988: 899). By contrast, the mines of Thuringia, Erzebirge, Carpathians and the Eastern Alps produced a total of 96 tons of silver in A.D. 1526–1535. By the close of the sixteenth century, output of European silver mines was but 'a mere tenth of the American bullion imports then registered at Seville' (Brading and Cross, 1972: 545). Had the ore bounties of European mines been exploited under the conditions that Spanish American mines were, history would be much different today.

4. Conclusions

Silver mining enterprises in the West had a primary impulse in the Asian demand for silver. They were monopolies that allowed Europe to mine its lodes, first, and later the lodes of the Americas, under the conditions of a silver-hungry world economy. They responded to, and in turn deepened, money and trade relations that expanded bullion demand locally, regionally and on an inter-continental level. The monopoly of the *consilium* of thirteenth-century Trent evolved in dialectical relationship with local and regional conditions. Situated in a strategic highway of trade, the *consilium* exploited the mines of Monte Calisio. The forms and functions of its business increased class differentiation between rich and poor miners, all nonetheless participating from economic relations that were increasingly monetized. The advantage of monopolies, however, came at a price. Not only the lord Bishop of Trent imposed taxes to the business, seeking to grab higher shares of mineral rent, but ferocious competition against other economic agents holding usufruct rights over other natural resources was possibly the general situation. Nonetheless, the *consilium* had the advantage of small size. Lacking the large, continental business that some Florentine mining and commodity merchant houses had in this period, such as the Frescobaldi, it navigated better the shocks inherent to bullion production, distribution and valorization. Its monopoly business allowed a strategic principality to exploit an internal resource frontier upon which the city expanded its money and trade relations with other economic agents of the first silver cycle.

Spanish American monopoly mining enterprises began as land grants that allowed individual holders to exploit conquered lands containing ores and Amerindian labor. Their money relations evolved in dialectical response to local geological and geographic conditions but linked to the increasing monetary needs of the Spanish Empire. Zacatecas and Potosi mines yielded astonishingly high amounts of bullion, sustained by exploitation of Amerindians. The mining monopolies in the colonies deepened the peripheralization of mining labor while increasingly linking it to the money and trade relations of the world economy. Imports of Africans (forced into slavery) increased dramatically from the 5000 slaves a year on average in A.D. 1500–1550, to 1490,000 in A.D. 1600–1700 (Kamen, 2003: 137). Thousands were sold in the slave-trading markets of the colonies and sent to silver mining districts. 'The continuing promise of America' was a in time became an ideological force that allowed miners to pull cheap labor from Europe, too, poor Europeans continued to migrate in large numbers to the New World as late as the early eighteenth century (Kamen, 2003: 130–131).

This article painted in broad strokes some of the most extraordinary money relations unleashed by mining monopolies of the West in a Sino-centered world economy. Much, however, remains unanswered. First, there is the question of the relation between mining monopolies and political power. The *consilium* of Trent competed against the lord Bishop for higher shares of mining profits but relied upon the laws written by the lord Bishop criminalizing acts of violence in Monte Calisio. The owners of *asientos de minas* in Zacatecas and Potosi found ways to violate the payment of the one-fifth owed to the Spanish Treasury, but many exploited their connections to the colonial bureaucracy in matters pertaining mercury markets. Both cases suggest that the relation between monopolies and power was dynamic and complex, and depended upon the stage of maturity of the mining monopoly, and that of the territorial power at stake. Did owners of mining fortunes aspire to join the world of politics? Did a climate of war against Islam, as the one that intermittently affected Mediterranean businesses in the thirteenth century, influence the political mentality of the *consilium* of Trent? How strong was the loyalty of the *consilium* to the ecclesiastical principality upon which they built their fortunes? Where did the loyalties of rich miners of Spanish America lie? Answers to these questions will require a thorough study on the dialectic of mining wealth and territorial power.

A second and more daunting question is about the relation between mining monopolies and warfare. Many decades ago, Armando Sapori argued that production of the necessities of life could only be organized along monopolies in the Latin West. Such was the case because large commercial houses were on the ascent under conditions of ferocious competition ingrained to feudalism (Sapori, 1947). Ferocious competition meant war, economic and otherwise. How did mining monopolies exercise warfare? How did they use force, 'the midwife of every old society pregnant with a new one,' (Marx, 1906: 824) to surrender all autonomous economic life around them to their wishes? How do mining monopolies, once exploiting the gains of war, survive and endure that new society they fostered by economic violence? Finding clues to these questions will require building a typology of mining monopolies that includes other business forms that emerged across Eurasia besides the corporation and the *asientos de minas* discussed here. Such a typology ought to reveal the relations of co-optation, theft, bribery, land usurpation and even massacres, that notable capitalist mining monopolies exercised in the hinterlands of the West, and elsewhere in more recent times.

A third line of inquiry relates to the relation between money, mining monopolies and empire. Did mining monopolies and money build the twin foundations of historical empires? Did forward and backward money linkages create speculative gains feeding imperial competition? What were the macro-economic conditions that put such speculative gains at the service of empire? These questions require a detailed exploration of the overlapping chronological realities of monopolies, the world economy, and empires. Such an exploration must necessarily consider the money relations triggered by the production of staple metallic commodities of empire, such as the functions of tin on behalf of England, and those of copper, on behalf of Portugal, favored certain geopolitical scenarios that led to rise and fall of Western empires.

The broad strokes presented in this article hopefully serve to justify the need to address the above questions. Mining historians can afford not to stay behind the trenches of a theory of history that forces a shortsighted view of the complex world that mining in the West created, long before the Industrial Revolution. Unless we travel through the mining fields of the past, armed with a theory and history that accounts for the centuries-long, dynamic evolution of the world economy, we will be missing much of the mining history we endeavor to grasp. Too much is at stake today, when mining has proven that ferocious monopolistic competition is a principle it cannot live without, aided by a world economy that refuses to alter its addictive consumption of the metallic veins of the Earth.

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